

Example 7.22

Write a function `xy = xytrianglererv(m)` that generates m sample pairs (X, Y) in Example 7.16.

In Example 7.16, we found that

$$f_X(x) = \begin{cases} 2x & 0 \leq x \leq 1, \\ 0 & \text{otherwise,} \end{cases} \quad f_{Y|X}(y|x) = \begin{cases} 1/x & 0 \leq y \leq x, \\ 0 & \text{otherwise.} \end{cases} \quad (7.73)$$

```
function xy = xytrianglererv(m);
x=sqrt(rand(m,1));
y=x.*rand(m,1);
xy=[x y];
```

For $0 \leq x \leq 1$, we have that $F_X(x) = x^2$. Using Theorem 6.5 to generate sample values of X , we define $u = F_X(x) = x^2$. Then, for $0 < u < 1$, $x = \sqrt{u}$. By Theorem 6.5, if U is uniform $(0, 1)$, then $\sqrt{U}$ has PDF $f_X(x)$. Next, we observe that

given $X = x_i$, Y is the uniform $(0, x_i)$ random variable. Given another uniform $(0, 1)$ random variable U_i , Theorem 6.3(a) states that $Y_i = x_i U_i$ is the uniform $(0, x_i)$ random variable. We implement these ideas in the function `xytrianglererv.m`.

Quiz 7.7

For random variables X and Y with joint PMF $P_{X,Y}(x, y)$ given in Example 7.11, write a MATLAB function `xy=dtrianglererv(m)` that generates m sample pairs.

Problems

Difficulty: ● Easy ■ Moderate ◆ Difficult ◆◆ Experts Only

7.1.1 ● Random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < -3, \\ 0.4 & -3 \leq x < 5, \\ 0.8 & 5 \leq x < 7, \\ 1 & x \geq 7. \end{cases}$$

Find the conditional CDF $F_{X|X>0}(x)$ and PMF $P_{X|X>0}(x)$.

7.1.2 ● X is the discrete uniform $(0, 5)$ random variable. What is $E[X|X \geq E[X]]$?

7.1.3 ● X has PMF

$$P_X(x) = \binom{4}{x} (1/2)^4.$$

Find $P_{X|B}(x)$ where $B = \{X \neq 0\}$.

7.1.4 ● In a youth basketball league, a player is fouled in the act of shooting a layup. There is a probability $q = 0.2$ that the layup is good, scoring 2 points. If the layup is good, the player is also awarded 1 free throw, giving the player a chance at a three-point play. If the layup is missed, then (because of the foul) the player is still awarded one point automatically and is also awarded one free throw, enabling a chance to score two points in total. The player makes a free throw with probability $p = 1/2$.

- What is the PMF of X , the number of points scored by the player?
- Find the conditional PMF $P_{X|T}(x)$ of X given event T that the free throw is good.

7.1.5 ■ Every day you consider going jogging. Before each mile, including the first, you will quit with probability q , independent of the number of miles you have already run. However, you are sufficiently decisive that you never run a fraction of a mile. Also, we say you have run a marathon whenever you run at least 26 miles.

- Let M equal the number of miles that you run on an arbitrary day. Find the PMF $P_M(m)$.
- Let r be the probability that you run a marathon on an arbitrary day. Find r .
- Let J be the number of days in one year (not a leap year) in which you run a marathon. Find the PMF $P_J(j)$. This answer may be expressed in terms of r found in part (b).
- Define $K = M - 26$. Let A be the event that you have run a marathon. Find $P_{K|A}(k)$.

7.1.6 ■ A random ECE student has height X in inches given by the PDF

$$f_X(x) = \frac{4e^{-(x-70)^2/8} + e^{-(x-65)^2/8}}{5\sqrt{8\pi}}.$$

- Sketch $f_X(x)$ over the interval $60 \leq x \leq 75$. (For purposes of sketching, note that $\sqrt{8\pi} \approx 5$.)
- Find the probability that a random ECE student is less than 5 feet 8 inches tall.
- Use conditional PDFs to explain why $f_X(x)$ might be a reasonable model for ECE students.

7.1.7 ■ A test for diabetes is a measurement X of a person's blood sugar level following an overnight fast. For a healthy person, a blood sugar level X in the range of 70–110 mg/dl is considered normal. When a measurement X is used as a test for diabetes, the result is called positive (event T^+) if $X \geq 140$; the test is negative (event T^-) if $X \leq 110$, and the test is ambiguous (event T^0) if $110 < X < 140$.

Given that a person is healthy (event H), a blood sugar measurement X is the Gaussian $(90, 20)$ random variable. Given that a person has diabetes, (event D), X is the Gaussian $(60, 40)$ random variable. A randomly chosen person is healthy with probability $P[H] = 0.9$ or has diabetes with probability $P[D] = 0.1$.

- What is the conditional PDF $f_{X|H}(x)$?
- Calculate the conditional probabilities $P[T^+|H]$, and $P[T^-|H]$.
- Find $P[H|T^-]$, the conditional probability that a person is healthy given the event of a negative test.
- When a person has an ambiguous test result (T^0), the test is repeated, possibly many times, until either a positive T^+ or negative T^- result is obtained. Let N denote the number of times the test is given. Assuming that for a given person the result of each test is independent of the result of all other tests, find the conditional PMF of N given event H that a person is healthy. Note that $N = 1$ if the person has a positive T^+ or negative T^- result on the first test.

7.1.8 ■ For the quantizer of Example 7.6, the difference $Z = X - Y$ is the quantization error or quantization “noise.” As in Example 7.6, assume that X has a uniform $(-r/2, r/2)$ PDF.

- Given event B_i that $Y = y_i = \Delta/2 + i\Delta$ and X is in the i th quantization interval, find the conditional PDF of Z .
- Show that Z is a uniform random variable. Find the PDF, the expected value, and the variance of Z .

7.1.9 ♦ For the quantizer of Example 7.6, we showed in Problem 7.1.8 that the quantization noise Z is a uniform random variable. If X is not uniform, show that Z is nonuniform by calculating the PDF of Z for a simple example.

7.2.1 ● X is the binomial $(5, 1/2)$ random variable. Find $P_{X|B}(x)$, where the condi-

tion $B = \{X \geq \mu_X\}$. What are $E[X|B]$ and $\text{Var}[X|B]$?

7.2.2 ● Random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ 0.2 & -1 \leq x < 0, \\ 0.7 & 0 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

Given $B = \{|X| > 0\}$, find $P_{X|B}(x)$. What are $E[X|B]$ and $\text{Var}[X|B]$?

7.2.3 ● X is the continuous uniform $(-5, 5)$ random variable. Given the event $B = \{|X| \leq 3\}$, find the

- conditional PDF, $f_{X|B}(x)$,
- conditional expected value, $E[X|B]$,
- conditional variance, $\text{Var}[X|B]$.

7.2.4 ● Y is the exponential (0.2) random variable. Given $A = \{Y < 2\}$, find:

- $f_{Y|A}(y)$,
- $E[Y|A]$.

7.2.5 ● For the experiment of spinning the pointer three times and observing the maximum pointer position, Example 4.5, find the conditional PDF given the event R that the maximum position is on the right side of the circle. What are the conditional expected value and the conditional variance?

7.2.6 ■ The number of pages X in a document has PMF

$$P_X(x) = \begin{cases} 0.15 & x = 1, 2, 3, 4, \\ 0.1 & x = 5, 6, 7, 8, \\ 0 & \text{otherwise.} \end{cases}$$

A firm sends all documents with an even number of pages to printer A and all documents with an odd number of pages to printer B .

- Find the conditional PMF of the length X of a document, given the document was sent to A . What are the conditional expected length and standard deviation?

- Find the conditional PMF of the length X of a document, given the document was sent to B and had no more than six pages. What are the conditional expected length and standard deviation?

7.2.7 ■ Select integrated circuits, test them in sequence until you find the first failure, and then stop. Let N be the number of tests. All tests are independent, with probability of failure $p = 0.1$. Consider the condition $B = \{N \geq 20\}$.

- Find the PMF $P_N(n)$.
- Find $P_{N|B}(n)$, the conditional PMF of N given that there have been 20 consecutive tests without a failure.
- What is $E[N|B]$, the expected number of tests given that there have been 20 consecutive tests without a failure?

7.2.8 ■ W is the Gaussian $(0, 4)$ random variable. Given the event $C = \{W > 0\}$, find the conditional PDF, $f_{W|C}(w)$, the conditional expected value, $E[W|C]$, and the conditional variance, $\text{Var}[W|C]$.

7.2.9 ■ The time between telephone calls at a telephone switch is the exponential random variable T with expected value 0.01.

- What is $E[T|T > 0.02]$, the conditional expected value of T ?
- What is $\text{Var}[T|T > 0.02]$, the conditional variance of T ?

7.2.10 ■ As the final rider in the final 60 km time trial of the Tour de France, Roy must finish in time $T \leq 1$ hour to win the Tour. He has the choice of bike made of (1) carbon fiber or (2) titanium. On the carbon fiber bike, his speed V over the course is the continuous uniform random variable with $E[V] = 58$ km/hr and $\text{Var}[V] = 12$. On the titanium bike, V is the exponential random variable with $E[V] = 60$ km/hr.

- Roy chooses his bike to maximize $P[W]$, the probability he wins the Tour. Which bike does Roy choose and what is $P[W]$?
- Suppose instead that Roy flips a fair coin to choose his bike. What is $P[W]$?

7.2.11 ■ For the distance D of a shot-put toss in Problem 4.7.8, find the conditional PDFs $f_{D|D>0}(d)$ and $f_{D|D\leq 70}(d)$.

7.3.1 ● X and Y are independent identical discrete uniform $(1, 10)$ random variables. Let A denote the event that $\min(X, Y) > 5$. Find the conditional PMF $P_{X,Y|A}(x, y)$.

7.3.2 ● Continuing Problem 7.3.1, let B denote the event that $\max(X, Y) \leq 5$. Find the conditional PMF $P_{X,Y|B}(x, y)$.

7.3.3 ● Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 6e^{-(2x+3y)} & x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Let A be the event that $X + Y \leq 1$. Find the conditional PDF $f_{X,Y|A}(x, y)$.

7.3.4 ■ N and K have joint PMF

$$P_{N,K}(n, k) = \begin{cases} \frac{(1-p)^{n-1}p}{n} & n=1, 2, \dots, \\ 0 & k=1, \dots, n, \\ & \text{otherwise.} \end{cases}$$

Let B denote the event that $N \geq 10$.

- Find the conditional PMFs $P_{N|B}(n)$ and $P_{N,K|B}(n, k)$. Which should you find first?
- Find the conditional expected values $E[N|B]$, $E[K|B]$, $E[N + K|B]$, $\text{Var}[N|B]$, $\text{Var}[K|B]$, $E[NK|B]$.

7.3.5 ■ X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} (x+y)/3 & 0 \leq x \leq 1; \\ & 0 \leq y \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

Let $A = \{Y \leq 1\}$.

- What is $P[A]$?
- Find $f_{X,Y|A}(x, y)$.
- Find $f_{X|A}(x)$ and $f_{Y|A}(y)$.

7.3.6 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} (4x + 2y)/3 & 0 \leq x \leq 1; \\ & 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $A = \{Y \leq 1/2\}$.

- What is $P[A]$?
- Find $f_{X,Y|A}(x, y)$.
- Find $f_{X|A}(x)$, and $f_{Y|A}(y)$.

7.3.7 ■ A study examined whether there was correlation between how much football a person watched and how bald the person was. The time T watching football was measured on a 0, 1, 2 scale such that $T = 0$ if a person never watched football, $T = 1$ if a person watched football occasionally, and $T = 2$ if a person watched a lot of football. Similarly, baldness B was measured on the same scale: $B = 0$ for a person with a full head of hair, $B = 1$ for a person with thinning hair, and $B = 2$ for a person who has not much hair at all. The experiment was to learn B and T for a randomly chosen person, equally likely to be a man (event M) or a woman (event W). The study found that given a person was a man (event M), random variables B and T were conditionally independent. Similarly, given that a person was a woman (event W), B and T were conditionally independent. Moreover, B and T had conditional joint PMFs

b	0	1	2
$P_{B M}(b)$	0.2	0.3	0.5
t	0	1	2
$P_{T M}(t)$	0.2	0.2	0.6
b	0	1	2
$P_{B W}(b)$	0.6	0.3	0.1
t	0	1	2
$P_{T W}(t)$	0.6	0.2	0.2

- Find the conditional PMF $P_{B,T|W}(b, t)$ of B and T given that a person is a woman.
- Find the conditional PMF $P_{B,T|M}(b, t)$ of B and T given that a person is a man.
- Find the joint PMF $P_{B,T}(b, t)$.
- Find the covariance of B and T . Are B and T independent?

7.3.8 ♦ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 5x^2/2 & -1 \leq x \leq 1; \\ & 0 \leq y \leq x^2, \\ 0 & \text{otherwise.} \end{cases}$$

Let $A = \{Y \leq 1/4\}$.

- Find the conditional PDF $f_{X,Y|A}(x,y)$.
- Find $f_{Y|A}(y)$ and $E[Y|A]$.
- Find $f_{X|A}(x)$ and $E[X|A]$.

7.3.9 ♦ X and Y are independent random variables with PDFs

$$f_X(x) = \begin{cases} 2x & 0 \leq x \leq 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$f_Y(y) = \begin{cases} 3y^2 & 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $A = \{X > Y\}$.

- What are $E[X]$ and $E[Y]$?
- What are $E[X|A]$ and $E[Y|A]$?

7.4.1 ● Given $X = x$,

- Y_1 is Gaussian with conditional expected value x and conditional variance 1.
- Y_2 is Gaussian with conditional expected value x and conditional variance x^2 .

Find the conditional PDFs $f_{Y_1|X}(y_1|x)$ and $f_{Y_2|X}(y_2|x)$.

7.4.2 ● X is the continuous uniform $(0,1)$ random variable. Given $X = x$, Y has a continuous uniform $(0,x)$ PDF. What is the joint PDF $f_{X,Y}(x,y)$? Sketch the region of the X,Y plane for which $f_{X,Y}(x,y) > 0$.

7.4.3 ● X is the continuous uniform $(0,1)$ random variable. Given $X = x$, Y is conditionally a continuous uniform $(0,1+x)$ random variable. What is the joint PDF $f_{X,Y}(x,y)$ of X and Y ?

7.4.4 ● Z is a Gaussian $(0,1)$ noise random variable that is independent of X , and

$Y = X + Z$ is a noisy observation of X . What is the conditional PDF $f_{Y|X}(y|x)$?

7.4.5 ● A business trip is equally likely to take 2, 3, or 4 days. After a d -day trip, the change in the traveler's weight, measured as an integer number of pounds, is a uniform $(-d,d)$ random variable. For one such trip, denote the number of days by D and the change in weight by W . Find the joint PMF $P_{D,W}(d,w)$.

7.4.6 ● X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} (4x+2y)/3 & 0 \leq x \leq 1; \\ & 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- For which values of y is $f_{X|Y}(x|y)$ defined? What is $f_{X|Y}(x|y)$?
- For which values of x is $f_{Y|X}(y|x)$ defined? What is $f_{Y|X}(y|x)$?

7.4.7 ● A student's final exam grade depends on how close the student sits to the center of the classroom during lectures. If a student sits r feet from the center of the room, the grade is a Gaussian random variable with expected value $80 - r$ and standard deviation r . If r is a sample value of random variable R , and X is the exam grade, what is $f_{X|R}(x|r)$?

7.4.8 ■ $Y = ZX$ where X is the Gaussian $(0,1)$ random variable and Z , independent of X , has PMF

$$P_Z(z) = \begin{cases} 1-p & z = -1, \\ p & z = 1. \end{cases}$$

True or False:

- Y and Z are independent.
- Y and X are independent.

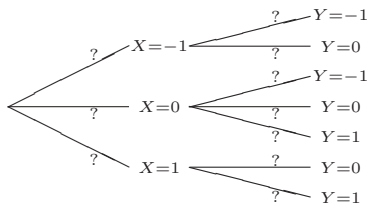
7.4.9 ■ At the One Top Pizza Shop, mushrooms are the only topping. Curiously, a pizza sold before noon has mushrooms with probability $p = 1/3$ while a pizza sold after noon never has mushrooms. Also, a pizza is equally likely to be sold before noon as

after noon. On a day in which 100 pizzas are sold, let N equal the number of pizzas sold before noon and let M equal the number of mushroom pizzas sold during the day. What is the joint PMF $P_{M,N}(m,n)$? Are M and N independent? Hint: Find the conditional PMF of M given N .

7.4.10 ■ Random variables X and Y have the joint PMF in the following table.

$P_{X,Y}(x,y)$	$y = -1$	$y = 0$	$y = 1$
$x = -1$	3/16	1/16	0
$x = 0$	1/6	1/6	1/6
$x = 1$	0	1/8	1/8

- Are X and Y independent?
- The experiment from which X and Y are derived is performed sequentially. First, X is found, then Y is found. In this context, label the conditional branch probabilities of the following tree:



7.4.11 ■ Flip a coin twice. On each flip, the probability of heads equals p . Let X_i equal the number of heads (either 0 or 1) on flip i . Let $W = X_1 - X_2$ and $Y = X_1 + X_2$. Find $P_{W,Y}(w,y)$, $P_{W|Y}(w|y)$, and $P_{Y|W}(y|w)$.

7.4.12 ■ Show that

$$\begin{aligned} \lim_{\Delta \rightarrow 0} \mathbb{P}[x_1 < X \leq x_2 | y < Y \leq y + \Delta] \\ = \int_{x_1}^{x_2} f_{X|Y}(x|y) \, dx. \end{aligned}$$

Hint: $\mathbb{P}[x_1 < X \leq x_2, y < Y \leq y + \Delta]$ can be written as an integral of $f_{X,Y}(x,y)$.

7.4.13 ♦ Packets arriving at an Internet router are either voice packets (v) or data packets (d). Each packet is a voice packet

with probability p , independent of any other packet. Observe packets at the Internet router until you see two voice packets. Let M equal the number of packets up to and including the first voice packet. Let N equal the number of packets observed up to and including the second voice packet. Find the conditional PMFs $P_{M|N}(m|n)$ and $P_{N|M}(n|m)$. Interpret your results.

7.4.14 ♦♦ Suppose you arrive at a bus stop at time 0, and at the end of each minute, with probability p , a bus arrives, or with probability $1 - p$, no bus arrives. Whenever a bus arrives, you board that bus with probability q and depart. Let T equal the number of minutes you stand at a bus stop. Let N be the number of buses that arrive while you wait at the bus stop.

- Identify the set of points (n,t) for which $\mathbb{P}[N = n, T = t] > 0$.
- Find $P_{N,T}(n,t)$.
- Find the marginal PMFs $P_N(n)$ and $P_T(t)$.
- Find the conditional PMFs $P_{N|T}(n|t)$ and $P_{T|N}(t|n)$.

7.4.15 ♦♦ Each millisecond at an Internet router, a packet independently arrives with probability p . Each packet is either a data packet (d) with probability q or a video packet (v). Each data packet belongs to an email with probability r . Let N equal the number of milliseconds required to observe the first 100 email packets. Let T equal the number of milliseconds you observe the router waiting for the first email packet. Find the marginal PMF $P_T(t)$ and the conditional PMF $P_{N|T}(n|t)$. Lastly, find the conditional PMF $P_{T|N}(t|n)$.

7.5.1 ● X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the PDF $f_Y(y)$, the conditional PDF $f_{X|Y}(x|y)$, and the conditional expected value $\mathbb{E}[X|Y = y]$.

7.5.2 Let random variables X and Y have joint PDF $f_{X,Y}(x, y)$ given in Problem 7.5.1. Find the PDF $f_X(x)$, the conditional PDF $f_{Y|X}(y|x)$, and the conditional expected value $E[Y|X = x]$.

7.5.3 The probability model for random variable A is

$$P_A(a) = \begin{cases} 1/3 & a = -1, \\ 2/3 & a = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The conditional probability model for random variable B given A is:

$$P_{B|A}(b|1) = \begin{cases} 1/3 & b = 0, \\ 2/3 & b = 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$P_{B|A}(b|1) = \begin{cases} 1/2 & b = 0, \\ 1/2 & b = 1, \\ 0 & \text{otherwise.} \end{cases}$$

- What is the probability model for random variables A and B ? Write the joint PMF $P_{A,B}(a, b)$ as a table.
- If $A = 1$, what is the conditional expected value $E[B|A = 1]$?
- If $B = 1$, what is the conditional PMF $P_{A|B}(a|1)$?
- If $B = 1$, what is the conditional variance $\text{Var}[A|B = 1]$ of A ?
- What is the covariance $\text{Cov}[A, B]$?

7.5.4 For random variables A and B given in Problem 7.5.3, let $U = E[B|A]$. Find the PMF $P_U(u)$. What is $E[U] = E[E[B|A]]$?

7.5.5 Random variables N and K have the joint PMF

$$P_{N,K}(n, k) = \begin{cases} \frac{100^n e^{-100}}{(n+1)!} & n=0, 1, \dots; \\ & k=0, 1, \dots, n, \\ 0 & \text{otherwise.} \end{cases}$$

- Find the marginal PMF $P_N(n)$ and the conditional PMF $P_{K|N}(k|n)$.
- Find the conditional expected value $E[K|N = n]$.

- Express the random variable $E[K|N]$ as a function of N and use the iterated expectation to find $E[K]$.

7.5.6 Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 1/2 & -1 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- What is $f_Y(y)$?
- What is $f_{X|Y}(x|y)$?
- What is $E[X|Y = y]$?

7.5.7 Over the circle $X^2 + Y^2 \leq r^2$, random variables X and Y have the uniform PDF

$$f_{X,Y}(x, y) = \begin{cases} 1/(\pi r^2) & x^2 + y^2 \leq r^2, \\ 0 & \text{otherwise.} \end{cases}$$

- What is $f_{Y|X}(y|x)$?
- What is $E[Y|X = x]$?

7.5.8 (Continuation of Problem 4.6.14) At time $t = 0$, the price of a stock is a constant k dollars. At time $t > 0$ the price of a stock is a Gaussian random variable X with $E[X] = k$ and $\text{Var}[X] = t$. At time t , a *Call Option at Strike k* has value

$$V = (X - k)^+,$$

where the operator $(\cdot)^+$ is defined as $(z)^+ = \max(z, 0)$. Suppose that at the start of each $t = 30$ day month, you can buy the call option at strike k at a price D that is a random variable that fluctuates every month. You decide to buy the call only if the price D is no more than a threshold d^* . What value of the threshold d^* maximizes the expected return $E[R]$?

7.5.9 In a weekly lottery, each \$1 ticket sold adds 50 cents to the jackpot that starts at \$1 million before any tickets are sold. The jackpot is announced each morning to encourage people to play. On the morning of the i th day before the drawing, the current value of the jackpot J_i is announced.

On that day, the number of tickets sold, N_i , is a Poisson random variable with expected value J_i . Thus, six days before the drawing, the morning jackpot starts at \$1 million and N_6 tickets are sold that day. On the day of the drawing, the announced jackpot is J_0 dollars and N_0 tickets are sold before the evening drawing. What are the expected value and variance of J , the value of the jackpot the instant before the drawing? Hint: Use conditional expectations.

7.6.1 ● You wish to measure random variable X with expected value $E[X] = 1$ and variance $\text{Var}[X] = 1$, but your measurement procedure yields the noisy observation $Y = X + Z$, where Z is the Gaussian $(0, 2)$ noise that is independent of X .

- Find the conditional PDF $f_{Z|X}(z|x)$ of Z given $X = x$.
- Find the conditional PDF $f_{Y|X}(y|2)$ of Y given $X = 2$. Hint: Given $X = x$, $Y = x + Z$.

7.6.2 ■ X and Y are jointly Gaussian random variables with $E[X] = E[Y] = 0$ and $\text{Var}[X] = \text{Var}[Y] = 1$. Furthermore, $E[Y|X] = X/2$. Find $f_{X,Y}(x, y)$.

7.6.3 ■ A study of bicycle riders found that a male cyclist's speed X (in miles per hour over a 100-mile "century" ride) and weight Y (kg) could be modeled by a bivariate Gaussian PDF $f_{X,Y}(x, y)$ with parameters $\mu_X = 20$, $\sigma_X = 2$, $\mu_Y = 75$, $\sigma_Y = 5$ and $\rho_{X,Y} = -0.6$. In addition, a female cyclist's speed X' and weight Y' could be modeled by a bivariate Gaussian PDF $f_{X',Y'}(x', y')$ with parameters $\mu_{X'} = 15$, $\sigma_{X'} = 2$, $\mu_{Y'} = 50$, $\sigma_{Y'} = 5$ and $\rho_{X',Y'} = -0.6$. For men and women, the negative correlation of speed and weight reflects the common wisdom that fast cyclists are thin. As it happens, cycling is much more popular among men than women; in a mixed group of cyclists, a cyclist is a male with probability $p = 0.80$.

You suspect it's OK to ignore the differences between men and women since for both groups, weight and speed are negatively correlated, with $\rho = -0.6$. To convince yourself this is OK, you decide to study the speed $\hat{X}$ and weight $\hat{Y}$ of a cyclist randomly chosen from a large mixed group of male and female cyclists. How are $\hat{X}$ and $\hat{Y}$ correlated? Explain your answer.

7.6.4 ♦ Let X_1 and X_2 have a bivariate Gaussian PDF with correlation coefficient ρ_{12} such that each X_i is a Gaussian (μ_i, σ_i) random variable. Show that $Y = X_1 X_2$ has variance

$$\begin{aligned} \text{Var}[Y] &= \sigma_1^2 \sigma_2^2 (1 + \rho_{12}^2) \\ &\quad + \sigma_1^2 \mu_2^2 + \mu_1^2 \sigma_2^2 - \mu_1^2 \mu_2^2. \end{aligned}$$

Hints: Look ahead to Problem 9.2.4 and also use the iterated expectation to find

$$E[X_1^2 X_2^2] = E[E[X_1^2 X_2^2 | X_2]].$$

7.6.5 ♦ Use the iterated expectation for a proof of Theorem 5.19 without integrals.

7.7.1 ● For the modem receiver voltage X with PDF given in Example 7.8, use MATLAB to plot the PDF and CDF of random variable X . Write a MATLAB function `x=modemrv(m)` that produces `m` samples of the modem voltage X .

7.7.2 ■ For the quantizer of Example 7.6, we showed in Problem 7.1.9 that the quantization noise Z is nonuniform if X is nonuniform. In this problem, we examine whether it is a reasonable approximation to model the quantization noise as uniform. Consider the special case of a Gaussian $(0, 1)$ random variable X passed through a uniform b -bit quantizer over the interval $(-r/2, r/2)$ with $r = 6$. Does a uniform approximation get better or worse as b increases? Write a MATLAB program to generate histograms for Z to answer this question.